

Course 5491

5 Credits: 4

Program core

Source-grounded

Digital System Design

- To design of various combinational digital circuits using logic gates

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5491 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5491.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	5491
Course title	Digital System Design
Semester	5 Credits: 4
Course category	Program core
Periods per week	4 (L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To design of various combinational digital circuits using logic gates
- To analyze and design synchronous Sequential circuits
- To analyze and design asynchronous Sequential circuits
- To understand the concept of Hazards and design of Hazard free Circuit

Course outcomes

CO	Official outcome	Study level
CO1	Design Various types of Combinational Circuits 15 Understanding Analyze and Design Synchronous Sequential	See official syllabus
CO2	15 Understanding Circuit Analyze and Design Asynchronous Sequential	See official syllabus
CO3	15 Understanding Circuit Understand the various types of Hazards and design	See official syllabus
CO4	13 Understanding Hazard free Circuit	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 3 3
CO2	CO2 3 3 3
CO3	CO3 3 3 3
CO4	CO4 3 3 2

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Design Various types of Combinational Circuits	4	As specified
Module 2	M2.01 Analyze the given Synchronous Circuit 4 Apply	3	As specified
Module 3	Design and Analyze Asynchronous Sequential Network	4	As specified
Module 4	Design Hazard free Circuit	4	As specified

MODULE 1

Design Various types of Combinational Circuits

This module is presented from the official syllabus scope for Digital System Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Design Various types of Combinational Circuits
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

Design of different combinational Circuits 5 Apply

Design of different combinational Circuits 5 Apply is an official topic in Module 1 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design of different combinational Circuits 5 Apply using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design of different combinational circuits 5 apply should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design of different combinational Circuits 5 Apply means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Design of different combinational Circuits 5

Apply definition/meaning . steps, application . Technical terms English-
.

Design combinational Circuits using Multiplexers 4 Apply and decoders

Design combinational Circuits using Multiplexers 4 Apply and decoders is an official topic in Module 1 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design combinational Circuits using Multiplexers 4 Apply and decoders using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design combinational circuits using multiplexers 4 apply and decoders should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design combinational Circuits using Multiplexers 4 Apply and decoders means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- Design combinational Circuits using Multiplexers 4 Apply and decoders
 definition/meaning
 . steps, application
 . Technical terms English- .

Explain Different types of Adders

Explain Different types of Adders is an official topic in Module 1 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Different types of Adders using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different types of adders should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Different types of Adders means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain Different types of Adders
definition/meaning
application Technical terms English-
.

Design of Boolean function using Multiplexers 4 Apply and decoders and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder - Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit Design and Analyze Clocked Synchronous Sequential Network

Design of Boolean function using Multiplexers 4 Apply and decoders and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder - Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit Design and Analyze Clocked Synchronous Sequential Network is an official topic in Module 1 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design of Boolean function using Multiplexers 4 Apply and decoders and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder - Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit Design and Analyze Clocked Synchronous Sequential Network using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design of boolean function using multiplexers 4 apply and decoders and full adders, magnitude comparator, decoder, encoder, priority encoder, multiplexer and demultiplexer, binary parallel adder - carry lookahead adder, bcd adder. combination logic design using multiplexer and decoder case study: digital trans-receiver / 8 bit arithmetic and logic unit design and analyze clocked synchronous

sequential network should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design of Boolean function using Multiplexers 4 Apply and decoders and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder – Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit Design and Analyze Clocked Synchronous Sequential Network means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- $\square\square$ Design of Boolean function using Multiplexers 4 Apply and decoders and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder – Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit Design and Analyze Clocked Synchronous Sequential Network $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Design Various types of Combinational Circuits

M1.01	Design of different combinational Circuits	5
Apply		

M1.02	Design combinational Circuits using Multiplexers	4
Apply		

M1.03	and decoders Explain Different types of Adders	2
Understanding		

M1.04	Design of Boolean function using Multiplexers	4
Apply		

and decoders
Contents: Problem formulation and design of combinational circuits - Code-Converters, Half and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder – Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder
Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit

Design and Analyze Clocked Synchronous Sequential Network

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

M2.01 Analyze the given Synchronous Circuit 4 Apply

This module is presented from the official syllabus scope for Digital System Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: M2.01 Analyze the given Synchronous Circuit 4 Apply
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Analyze the given Synchronous Circuit 4 Apply Design of Clocked Synchronous Sequential 4 Apply M2.02 Circuit

Analyze the given Synchronous Circuit 4 Apply Design of Clocked Synchronous Sequential 4 Apply M2.02 Circuit is an official topic in Module 2 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Analyze the given Synchronous Circuit 4 Apply Design of Clocked Synchronous Sequential 4 Apply M2.02 Circuit using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, analyze the given synchronous circuit 4 apply design of clocked synchronous sequential 4 apply m2.02 circuit should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Analyze the given Synchronous Circuit 4 Apply Design of Clocked Synchronous Sequential 4 Apply M2.02 Circuit means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓ Analyze the given Synchronous Circuit 4 Apply Design of Clocked Synchronous Sequential 4 Apply M2.02 Circuit ഓഓഓഓഓഓഓഓഓഓ definition/meaning ഓഓഓഓഓഓഓഓഓഓ. ഓഓഓഓഓ ഓഓഓഓഓ ഓഓഓഓഓ, steps, application ഓഓഓഓഓ ഓഓഓഓഓഓഓഓ ഓഓഓഓഓ. Technical terms English-ഓ ഓഓഓഓഓ ഓഓഓഓഓഓഓഓഓഓ.

Design Various types of counters 4 Apply

Design Various types of counters 4 Apply is an official topic in Module 2 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design Various types of counters 4 Apply using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design various types of counters 4 apply should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design Various types of counters 4 Apply means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓർക്കൽ Design Various types of counters 4 Apply
ഓർക്കൽ definition/meaning ഓർക്കൽ. ഓർക്കൽ ഓർക്കൽ, steps, application ഓർക്കൽ ഓർക്കൽ. Technical terms English-ഓർക്കൽ
ഓർക്കൽ.

Design Various types of Registers 3 Apply Contents : Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF, Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked sequential circuits – Moore/Mealy models, state minimization, state assignment. Circuit implementation – Counters, Ripple Counters, Sequence Generator, Ring Counters, Shift registers, Universal Shift Register. Model Development: Designing of rolling display/real time clock

Design Various types of Registers 3 Apply Contents : Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF, Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked sequential circuits – Moore/Mealy models, state minimization, state assignment. Circuit implementation – Counters, Ripple Counters, Sequence Generator, Ring Counters, Shift registers, Universal Shift Register. Model Development: Designing of rolling display/real time clock is an official topic in Module 2 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Design Various types of Registers 3 Apply Contents : Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF, Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked sequential circuits – Moore/Mealy models, state minimization, state assignment. Circuit implementation – Counters, Ripple Counters, Sequence Generator, Ring Counters, Shift registers, Universal Shift Register. Model Development: Designing of rolling display/real time clock using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, design various types of registers 3 apply contents : flip flops – sr, jk, t, d, master/slave ff, triggering of ff, clocked synchronous sequential network

(cssn) -analysis and design of clocked sequential circuits – moore/mealy models, state minimization, state assignment. circuit implementation - counters, ripple counters, sequence genrator, ring counters, shift registers, universal shift register. model development: designing of rolling display/real time clock should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Design Various types of Registers 3 Apply Contents : Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF, Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked sequential circuits – Moore/Mealy models, state minimization, state assignment. Circuit implementation - Counters, Ripple Counters, Sequence Genrator, Ring Counters, Shift registers, Universal Shift Register. Model Development: Designing of rolling display/real time clock means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നു Design Various types of Registers 3 Apply Contents : Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF, Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked sequential circuits – Moore/Mealy models, state minimization, state assignment. Circuit implementation - Counters, Ripple Counters, Sequence Genrator, Ring Counters, Shift registers, Universal Shift Register. Model Development: Designing of rolling display/real time clock എന്നർത്ഥം definition/meaning എന്നർത്ഥം. എന്നർത്ഥം എന്നർത്ഥം, steps, application എന്നർത്ഥം എന്നർത്ഥം എന്നർത്ഥം. Technical terms English-എന്നു എന്നർത്ഥം എന്നർത്ഥം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

M2.01 Apply	Analyze the given Synchronous Circuit	4
Apply M2.02	Design of Clocked Synchronous Sequential Circuit	4
M2.03 Apply	Design Various types of counters	4
M2.04 Apply	Design Various types of Registers	3
	Series Test 1	1

Contents :

Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF,
Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked
sequential circuits – Moore/Mealy models, state minimization, state assignment.
Circuit implementation - Counters, Ripple Counters, Sequence Generator, Ring
Counters,
Shift registers, Universal Shift Register.
Model Development: Designing of rolling display/real time clock

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Design and Analyze Asynchronous Sequential Network

This module is presented from the official syllabus scope for Digital System Design.
Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Design and Analyze Asynchronous Sequential Network
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Analyze the given Asynchronous Circuit 4 Apply Explain stable state and unstable state and flow

Analyze the given Asynchronous Circuit 4 Apply Explain stable state and unstable state and flow is an official topic in Module 3 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Analyze the given Asynchronous Circuit 4 Apply Explain stable state and unstable state and flow using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, analyze the given asynchronous circuit 4 apply explain stable state and unstable state and flow should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Analyze the given Asynchronous Circuit 4 Apply Explain stable state and unstable state and flow means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓഫ് Analyze the given Asynchronous Circuit 4 Apply Explain stable state and unstable state and flow ഓഫ് ഓഫ് ഓഫ് definition/meaning ഓഫ് ഓഫ് ഓഫ്. ഓഫ് ഓഫ് ഓഫ്, steps, application ഓഫ് ഓഫ് ഓഫ്. Technical terms English-ഓഫ് ഓഫ് ഓഫ്.

4 Understanding table with example

4 Understanding table with example is an official topic in Module 3 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding table with example using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding table with example should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding table with example means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Understanding table with example
പരമ്പരാഗതമായി പട്ടിക definition/meaning പരമ്പരാഗതമായി. പട്ടികയിൽ പട്ടികയിൽ പട്ടികയിൽ,
steps, application പട്ടികയിൽ പട്ടികയിൽ പട്ടികയിൽ. Technical terms English-ൽ പട്ടികയിൽ
പട്ടികയിൽ പട്ടികയിൽ.

Explain State Assignment

Explain State Assignment is an official topic in Module 3 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain State Assignment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain state assignment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain State Assignment means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain State Assignment $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square\square$.

Explain races and cycles in ASC 3 Understanding Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC

Explain races and cycles in ASC 3 Understanding Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC is an official topic in Module 3 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain races and cycles in ASC 3 Understanding Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain races and cycles in asc 3 understanding analysis of asynchronous sequential circuits (asc), flow table reduction- state assignment problem and the transition table, stable state, unstable state, races and cycles in asc should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain races and cycles in ASC 3 Understanding Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain races and cycles in ASC 3
 Understanding Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC definition/meaning, steps, application. Technical terms English- Malayalam.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Design and Analyze Asynchronous Sequential Network

M3.01	Analyze the given Asynchronous Circuit	4
Apply		
	Explain stable state and unstable state and flow	
M3.02		4
Understanding	table with example	
	Explain State Assignment	
M3.03		4
Understanding		
	Explain races and cycles in ASC	
M3.04		3
Understanding		

Contents:

Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Design Hazard free Circuit

This module is presented from the official syllabus scope for Digital System Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Design Hazard free Circuit
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain Data synchronizer

Explain Data synchronizer is an official topic in Module 4 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Data synchronizer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain data synchronizer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Data synchronizer means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain Data synchronizer definition/meaning . steps, application . Technical terms English-

Explain Static and Dynamic Hazards

Explain Static and Dynamic Hazards is an official topic in Module 4 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Static and Dynamic Hazards using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain static and dynamic hazards should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Static and Dynamic Hazards means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain Static and Dynamic Hazards

Explain Static and Dynamic Hazards. definition/meaning, steps, application. Technical terms English- Malayalam.

Explain Essential Hazards

Explain Essential Hazards is an official topic in Module 4 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Essential Hazards using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain essential hazards should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Essential Hazards means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നു പറയുന്ന Explain Essential Hazards എന്നതിന്റെ definition/meaning എന്താണത്. എന്നതിന്റെ steps, application എന്നിവയെക്കുറിച്ചും എഴുതുക. Technical terms English-ൽ എഴുതുക.

Explain MOM and Practical Issues 4 Understanding Contents : Hazards: Static Hazards, Dynamic Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter, Synchronous and asynchronous inputs - switch bouncing Text/Reference: T/R Book Title/Author T1 Digital Principles and Design, Donald D. Givone, McGraw Hill Higher Education, 2022 T2 Modern Digital Electronics - R. P. Jain, 3rd Edition, 2007- Tata McGraw-Hill R1 Digital Design- Morris Mano, PHI, 4th Edition,2006 R2 Charles H. Roth, Jr, 'Fundamentals of Logic Design', Jaico Books, 4th Edition, 2002. R3 William I. Fletcher, "An Engineering Approach to Digital Design", Prentice-Hall of India, 1980. R4 John. F. Wakerly, "Digital Design Principles and Practices", Pearson Education, 4th Edition,2007. Online Resources: Sl. No. Website Link 1 https://onlinecourses.nptel.ac.in/noc21_ee39 2 https://onlinecourses.nptel.ac.in/noc25_ee24/preview 3 <https://www.geeksforgeeks.org/principles-in-digital-system-design/>

Explain MOM and Practical Issues 4 Understanding Contents : Hazards: Static Hazards, Dynamic Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter, Synchronous and asynchronous inputs - switch bouncing Text/Reference: T/R Book Title/Author T1 Digital Principles and Design, Donald D. Givone, McGraw Hill Higher Education, 2022 T2 Modern Digital Electronics - R. P. Jain, 3rd Edition, 2007- Tata McGraw-Hill R1 Digital Design- Morris Mano, PHI, 4th Edition,2006 R2 Charles H. Roth, Jr, 'Fundamentals of Logic Design', Jaico Books, 4th Edition, 2002. R3 William I. Fletcher, "An Engineering Approach to Digital Design", Prentice-Hall of India, 1980. R4 John. F. Wakerly, "Digital Design Principles and Practices", Pearson Education, 4th Edition,2007. Online Resources: Sl. No. Website Link 1 https://onlinecourses.nptel.ac.in/noc21_ee39 2 https://onlinecourses.nptel.ac.in/noc25_ee24/preview 3 <https://www.geeksforgeeks.org/principles-in-digital-system-design/> is an official topic in Module 4 of Digital System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain MOM and Practical Issues 4 Understanding Contents : Hazards: Static Hazards, Dynamic Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter, Synchronous and asynchronous inputs – switch bouncing Text/Reference: T/R Book Title/Author T1 Digital Principles and Design, Donald D. Givone, McGraw Hill Higher Education, 2022 T2 Modern Digital Electronics – R. P. Jain, 3rd Edition, 2007- Tata McGraw-Hill R1 Digital Design- Morris Mano, PHI, 4th Edition, 2006 R2 Charles H. Roth, Jr, 'Fundamentals of Logic Design', Jaico Books, 4th Edition, 2002. R3 William I. Fletcher, "An Engineering Approach to Digital Design", Prentice-Hall of India, 1980. R4 John. F. Wakerly, "Digital Design Principles and Practices", Pearson Education, 4th Edition, 2007. Online Resources: Sl. No. Website Link 1 https://onlinecourses.nptel.ac.in/noc21_ee39 2 https://onlinecourses.nptel.ac.in/noc25_ee24/preview 3 <https://www.geeksforgeeks.org/principles-in-digital-system-design/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain mom and practical issues 4 understanding contents : hazards: static hazards, dynamic hazards, essential hazards, pulse mode sequential circuits, design of hazard free circuits, data synchronizer mixed operating mode asynchronous circuits, practical issues- clock skew and jitter, synchronous and asynchronous inputs – switch bouncing text/reference: t/r book title/author t1 digital principles and design, donald d. givone, mcgraw hill higher education, 2022 t2 modern digital electronics – r. p. jain, 3rd edition, 2007- tata mcgraw-hill r1 digital design- morris mano, phi, 4th edition, 2006 r2 charles h. roth, jr, 'fundamentals of logic design', jaico books, 4th edition, 2002. r3 william i. fletcher, "an engineering approach to digital design", prentice-hall of india, 1980. r4 john. f. wakerly, "digital design principles and practices", pearson education, 4th edition, 2007. online resources: sl. no. website link 1 https://onlinecourses.nptel.ac.in/noc21_ee39 2 https://onlinecourses.nptel.ac.in/noc25_ee24/preview 3 <https://www.geeksforgeeks.org/principles-in-digital-system-design/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain MOM and Practical Issues 4 Understanding Contents : Hazards: Static Hazards, Dynamic Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter, Synchronous and asynchronous inputs – switch bouncing Text/Reference: T/R Book Title/Author T1 Digital Principles and Design, Donald D. Givone, McGraw Hill Higher Education, 2022 T2 Modern Digital Electronics – R. P. Jain, 3rd Edition, 2007- Tata McGraw-Hill R1 Digital Design- Morris Mano, PHI, 4th Edition,2006 R2 Charles H. Roth, Jr, 'Fundamentals of Logic Design', Jaico Books, 4th Edition, 2002. R3 William I. Fletcher, "An Engineering Approach to Digital Design", Prentice-Hall of India, 1980. R4 John. F. Wakerly, "Digital Design Principles and Practices", Pearson Education, 4th Edition,2007. Online Resources: Sl. No. Website Link 1 https://onlinecourses.nptel.ac.in/noc21_ee39 2 https://onlinecourses.nptel.ac.in/noc25_ee24/preview 3 https://www.geeksforgeeks.org/principles-in-digital-system-design/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain MOM and Practical Issues 4 Understanding Contents : Hazards: Static Hazards, Dynamic Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter, Synchronous and asynchronous inputs – switch bouncing Text/Reference: T/R Book Title/Author T1 Digital Principles and Design, Donald D. Givone, McGraw Hill Higher Education, 2022 T2 Modern Digital Electronics – R. P. Jain, 3rd Edition, 2007- Tata McGraw-Hill R1 Digital Design- Morris Mano, PHI, 4th Edition,2006 R2 Charles H. Roth, Jr, 'Fundamentals of Logic Design', Jaico Books, 4th Edition, 2002. R3 William I. Fletcher, "An Engineering Approach to Digital Design", Prentice-Hall of India, 1980. R4 John. F. Wakerly, "Digital Design Principles and Practices", Pearson Education, 4th Edition,2007. Online Resources: Sl. No. Website Link 1 https://onlinecourses.nptel.ac.in/noc21_ee39 2 https://onlinecourses.nptel.ac.in/noc25_ee24/preview 3 <https://www.geeksforgeeks.org/principles-in-digital-system-design/> definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Design Hazard free Circuit

M4.01	Explain Data synchronizer Understanding	3
M4.02	Explain Static and Dynamic Hazards Understanding	3
M4.03	Explain Essential Hazards Understanding	3
M4.04	Explain MOM and Practical Issues Understanding	4
	Series Test 2	1

Contents :

Hazards: Static Hazards, Dynamic Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer
Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter,
Synchronous and asynchronous inputs – switch bouncing

Text/Reference:

T/R

Book Title/Author

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Design of different combinational Circuits 5 Apply. (3 marks)

Answer: Begin with the standard definition or identity of *Design of different combinational Circuits 5 Apply*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Design combinational Circuits using Multiplexers 4 Apply and decoders. (3 marks)

Answer: Begin with the standard definition or identity of *Design combinational Circuits using Multiplexers 4 Apply and decoders*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain Different types of Adders. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Different types of Adders*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Design of Boolean function using Multiplexers 4 Apply and decoders and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder - Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit Design and Analyze Clocked Synchronous Sequential Network. (3 marks)

Answer: Begin with the standard definition or identity of *Design of Boolean function using Multiplexers 4 Apply and decoders and Full Adders, Magnitude Comparator, Decoder, Encoder, Priority Encoder, Multiplexer and Demultiplexer, Binary Parallel Adder - Carry lookahead Adder, BCD Adder. Combination logic design using Multiplexer and Decoder Case study: Digital trans-receiver / 8 bit Arithmetic and logic unit Design and Analyze Clocked Synchronous Sequential Network*. State the governing principle, list the

key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain Analyze the given Synchronous Circuit 4 Apply Design of Clocked Synchronous Sequential 4 Apply M2.02 Circuit. (3 marks)

Answer: Begin with the standard definition or identity of *Analyze the given Synchronous Circuit 4 Apply Design of Clocked Synchronous Sequential 4 Apply M2.02 Circuit*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Design Various types of counters 4 Apply. (3 marks)

Answer: Begin with the standard definition or identity of *Design Various types of counters 4 Apply*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Design Various types of Registers 3 Apply Contents : Flip flops - SR, JK, T, D, Master/Slave FF, Triggering of FF, Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked sequential circuits - Moore/Mealy models, state minimization, state assignment. Circuit implementation - Counters, Ripple Counters, Sequence Generator, Ring Counters, Shift registers, Universal Shift Register. Model Development: Designing of rolling display/real time clock. (3 marks)

Answer: Begin with the standard definition or identity of *Design Various types of Registers 3 Apply Contents : Flip flops – SR, JK, T, D, Master/Slave FF, Triggering of FF, Clocked Synchronous Sequential Network (CSSN) -Analysis and design of clocked sequential circuits – Moore/Mealy models, state minimization, state assignment. Circuit implementation - Counters, Ripple Counters, Sequence Generator, Ring Counters, Shift registers, Universal Shift Register. Model Development: Designing of rolling display/real time clock.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Analyze the given Asynchronous Circuit 4 Apply Explain stable state and unstable state and flow. (3 marks)

Answer: Begin with the standard definition or identity of *Analyze the given Asynchronous Circuit 4 Apply Explain stable state and unstable state and flow.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding table with example. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding table with example.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain State Assignment. (3 marks)

Answer: Begin with the standard definition or identity of *Explain State Assignment*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain races and cycles in ASC 3 Understanding Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC. (3 marks)

Answer: Begin with the standard definition or identity of *Explain races and cycles in ASC 3 Understanding Analysis of Asynchronous Sequential Circuits (ASC), Flow table reduction- State assignment problem and the transition table, Stable State, Unstable state, Races and cycles in ASC*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Explain Data synchronizer. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Data synchronizer*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain Static and Dynamic Hazards. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Static and Dynamic Hazards*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain Essential Hazards. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Essential Hazards*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain MOM and Practical Issues 4 Understanding Contents : Hazards: Static Hazards, Dynamic Hazards, Essential Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter, Synchronous and asynchronous inputs - switch bouncing Text/Reference: T/R Book Title/Author T1 Digital Principles and Design, Donald D. Givone, McGraw Hill Higher Education, 2022 T2 Modern Digital Electronics - R. P. Jain, 3rd Edition, 2007- Tata McGraw-Hill R1 Digital Design- Morris Mano, PHI, 4th Edition, 2006 R2 Charles H. Roth, Jr, 'Fundamentals of Logic Design', Jaico Books, 4th Edition, 2002. R3 William I. Fletcher, "An Engineering Approach to Digital Design", Prentice-Hall of India, 1980. R4 John. F. Wakerly, "Digital Design Principles and Practices", Pearson Education, 4th Edition, 2007. Online Resources: Sl. No. Website Link 1 https://onlinecourses.nptel.ac.in/noc21_ee39 2 https://onlinecourses.nptel.ac.in/noc25_ee24/preview 3 <https://www.geeksforgeeks.org/principles-in-digital-system-design/>. (3 marks)

Answer: Begin with the standard definition or identity of *Explain MOM and Practical Issues 4 Understanding Contents : Hazards: Static Hazards, Dynamic Hazards, Essential*

Hazards, Pulse mode sequential circuits, Design of Hazard free circuits, Data synchronizer Mixed operating mode asynchronous circuits, Practical issues- clock skew and jitter, Synchronous and asynchronous inputs – switch bouncing Text/Reference: T/R Book Title/Author T1 Digital Principles and Design, Donald D. Givone, McGraw Hill Higher Education, 2022 T2 Modern Digital Electronics – R. P. Jain, 3rd Edition, 2007- Tata McGraw-Hill R1 Digital Design- Morris Mano, PHI, 4th Edition, 2006 R2 Charles H. Roth, Jr, 'Fundamentals of Logic Design', Jaico Books, 4th Edition, 2002. R3 William I. Fletcher, "An Engineering Approach to Digital Design", Prentice-Hall of India, 1980. R4 John. F. Wakerly, "Digital Design Principles and Practices", Pearson Education, 4th Edition, 2007. Online Resources: Sl. No. Website Link 1 https://onlinecourses.nptel.ac.in/noc21_ee39 2 https://onlinecourses.nptel.ac.in/noc25_ee24/preview 3 <https://www.geeksforgeeks.org/principles-in-digital-system-design/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□ principle/steps, □□□□□ application □□□□ □□□□□□□□ □□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Design of different combinational Circuits**
5 Apply.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Analyze the given Synchronous Circuit 4 Apply Design of Clocked Synchronous Sequential 4 Apply M2.02 Circuit.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Analyze the given Asynchronous Circuit 4 Apply Explain stable state and unstable state and flow.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain Data synchronizer.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link https://onlinecourses.nptel.ac.in/noc21_ee39
- https://onlinecourses.nptel.ac.in/noc25_ee24/preview
- <https://www.geeksforgeeks.org/principles-in-digital-system-design/>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5491. Verify institutional updates against the current official curriculum.